AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #11

Activity Name: UART Programming

Name of Student:	**Agneay B Nair**
Registration No.:	**CH.SC.U4CSE24102**
Activity Date:	**21-08-26**
Submission Date:	**21-08-26**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

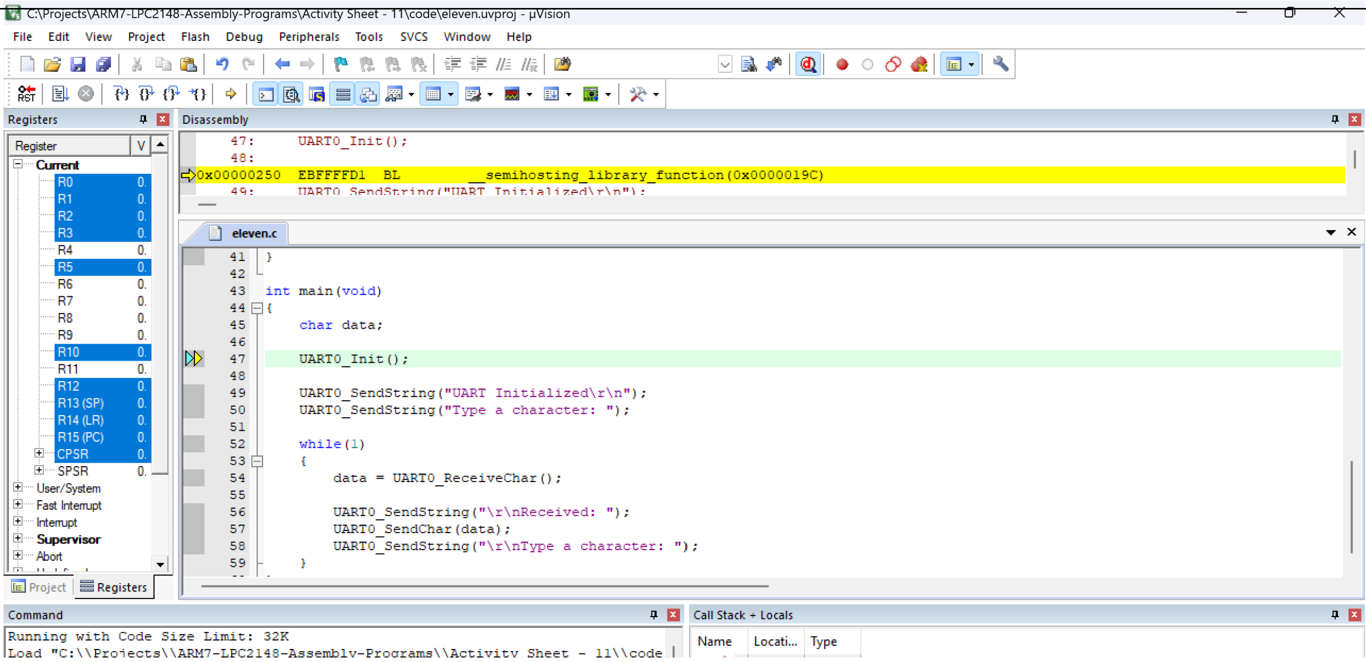
Write an embedded C program for ARM architecture to drive UART module. Debug and verify the program using IDE on a development computer system.

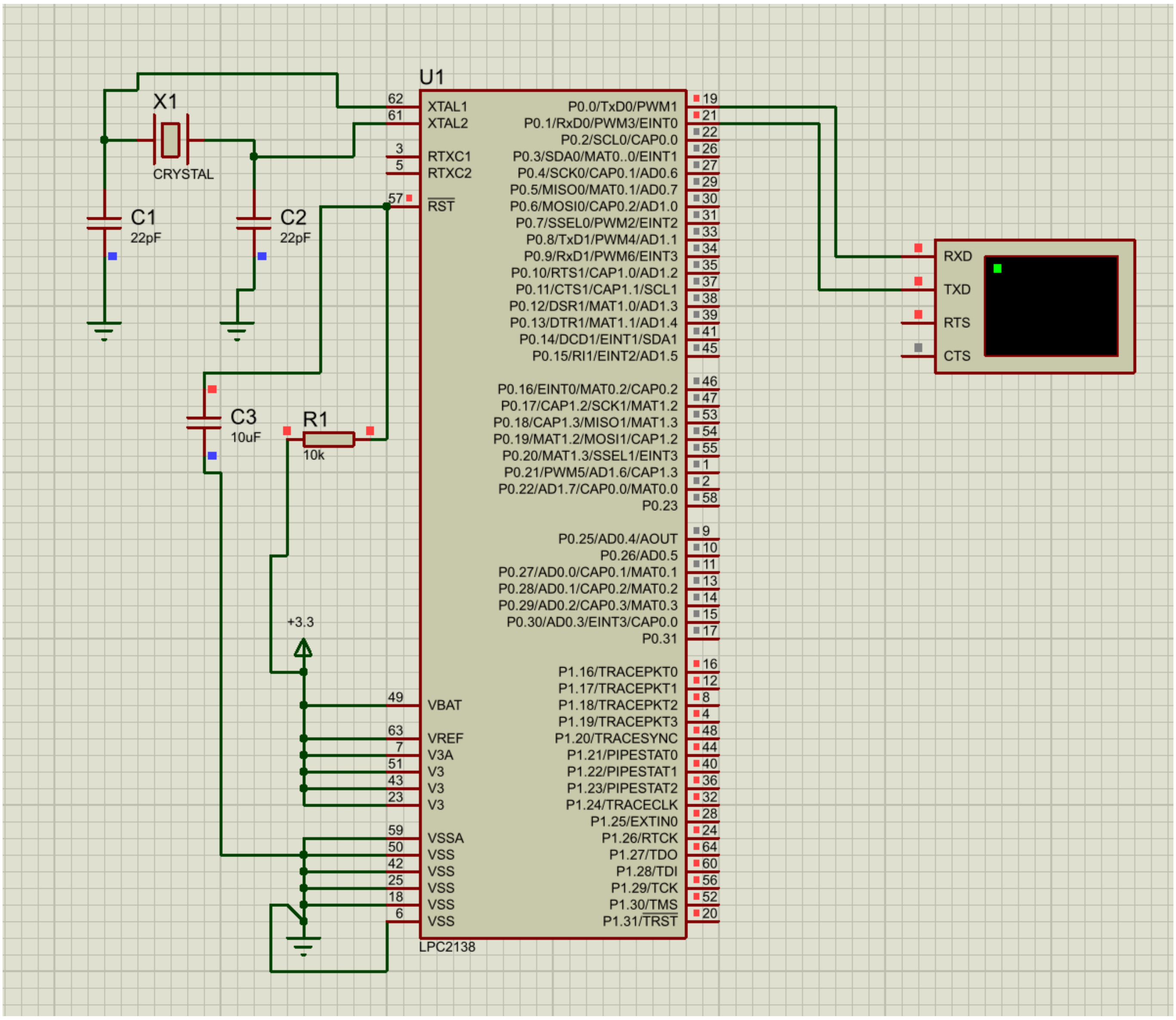
CODE:

```
#include <LPC214x.h>

void UART0_Init(void);
void UART0_SendChar(char ch);
void UART0_SendString(char *str);
char UART0_ReceiveChar(void);

void UART0_Init(void)
{
   // Select UART0 function
   // P0.0 = TXD0
   // P0.1 = RXD0
   PINSEL0 = 0x00000005;

   // Set Peripheral Clock = CCLK = 12 MHz
   VPBDIV = 0x01;

   // 8-bit data, 1 stop bit, no parity
   // Enable DLAB to set baud rate
   U0LCR = 0x83;

   // Set baud rate = 9600
   // PCLK = 12 MHz
   U0DLL = 78;
   U0DLM = 0;

   // Disable DLAB
   U0LCR = 0x03;
}

void UART0_SendChar(char ch)
{
   // Wait until transmitter is ready
   while (!(U0LSR & 0x20));

   // Send character
   U0THR = ch;
}

void UART0_SendString(char *str)
{
   while (*str)
   {
      UART0_SendChar(*str);
      str++;
```

```
    }
}

char UART0_ReceiveChar(void)
{
    // Wait until data is received
    while (!(U0LSR & 0x01));

    // Return received character
    return U0RBR;
}

int main(void)
{
    char data;

    // Initialize UART0
    UART0_Init();

    // Initial message
    UART0_SendString("UART Initialized\r\n");
    UART0_SendString("Type a character: ");

    while (1)
    {
        // Receive character
        data = UART0_ReceiveChar();

        // Display received character
        UART0_SendString("\r\nReceived: ");
        UART0_SendChar(data);

        // Ask for another character
        UART0_SendString("\r\nType a character: ");
    }
}
```

OUTPUT:


C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 11\code\eleven.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Disassembly
47: UART0_Init();
48:
0x00000250 EBFFFFD1 BL __semihosting_library_function(0x0000019C)
eleven.c
43 int main(void)
44 {
45 char data;
47 UART0_Init();
49 UART0_SendString("UART Initialized\r\n");
50 UART0_SendString("Type a character: ");
52 while(1)
53 {
54 data = UART0_ReceiveChar();
56 UART0_SendString("\r\nReceived: ");
57 UART0_SendChar(data);
58 UART0_SendString("\r\nType a character: ");
59 }
Command
Running with Code Size Limit: 32K
Call Stack + Locals
Name Locati... Type


U1
X1
CRYSTAL
C1
22pF
C2
22pF
C3
10uF
R1
10k
+3.3
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
VSSA
VSS
LPC2138
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
RXD
TXD
RTS
CTS

```
Type a character:
Received: A
Type a character:
```